

RENANCE

FEDERAL UNIVERSITY OF TECHNOLOGY, AKURE

MTH 104 — Past Questions & Materials

by Renance

Course: MTH 104

Institution: Federal University of Technology, Akure

Sources in this pack: 5

Includes past questions and lecture notes

This study pack was compiled and restructured by Renance for MTH 104 students at Federal University of Technology, Akure. It brings together 5 past paper(s)/material(s) into one clean, readable document so you can revise without chasing scattered PDFs. Practise these same questions interactively on Renance with instant answers, AI explanations, and full CBT mock exams — over 235,000+ questions across JAMB, Post-UTME, WAEC, NECO and university courses.

Study online: renance.app

Get the app: Google Play — [dev.renance.renance](https://play.google.com/store/apps/details?id=dev.renance.renance)

Generated by Renance AI

Paper 1 of 5 — Cyssa hub100LVL materials2nd SemesterMTH 104Ellipse and Hyperbola MTH 104.pdf

Lecture/Notes · Google Drive harvest

Do not write in this margin

Ellipse

QUESTION NO.

$$\frac{(x-h)^2}{a^2} + \frac{(y-k)^2}{b^2} = 1$$

$a > b$ Therefore major axis lies on y -axis
 Centre $C(h, k)$
 $c^2 = a^2 - b^2$
 Vertices are $V(h, k \pm a)$
 Focal, $f(h, k \pm c)$

$$\frac{(x-h)^2}{a^2} + \frac{(y-k)^2}{b^2} = 1$$

$a > b$ Therefore major axis lies on x -axis
 Centre $C(h, k)$
 $c^2 = a^2 - b^2$
 Vertices are $V(h \pm a, k)$
 Focal, $f(h \pm c, k)$

Do not write in this margin

Huneeba

Do not
write in
this margin

Ellipse

QUESTION NO.

Do not
write in
this margin

Standard Equation

$$\frac{(x-h)^2}{a^2} + \frac{(y-k)^2}{b^2} = 1, \quad a > b \text{ Major axis lies on } x\text{-axis}$$

Compare with Equation of circle

$$(x-h)^2 + (y-k)^2 = r^2$$

Centre $C(h, k)$

Example

$$\frac{x^2}{25} + \frac{y^2}{49} = 1$$

Since a is under y , major axis is along y -axis
 Since $a > b$ Centre $C(0, 0)$ There are 4 vertices

$$\therefore a^2 = 49 \Rightarrow a = 7$$

$$b^2 = 25 \Rightarrow b = 5$$

Focal, $F(h, k \pm c)$ Major axis, $MA = 2a = 14$ Minor axis $= 2b = 10$ end points Vertices are $= V(h, k \pm a) \Rightarrow V(0, \pm 7)$ $(0, 7)$ & $(0, -7)$ i.e. $V(0, 0 \pm 7)$

Focal is along major axis

b/w 0 & 7

$$c^2 = a^2 - b^2$$

$$c^2 = 49 - 25$$

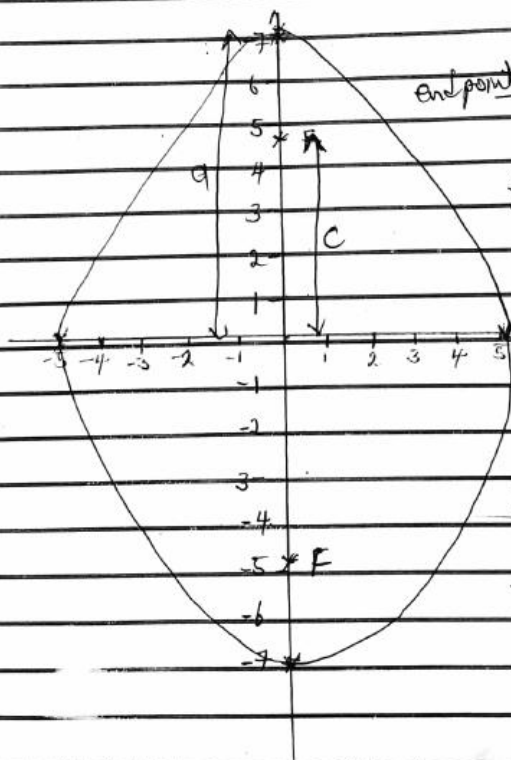
$$c = \sqrt{24} = 2\sqrt{6}$$

Focal $= (0, \pm 2\sqrt{6})$ i.e. $(0, \pm 4.8)$

$$= (0, \pm 4.8)$$

 $F(h, k \pm c)$

$$= (0, 4.8) \text{ \& } (0, -4.8)$$



Do not
write in
this marginDo not
write in
this margin

QUESTION NO.

Ex 1) $\frac{(x-2)^2}{16} + \frac{(y+3)^2}{9} = 1$

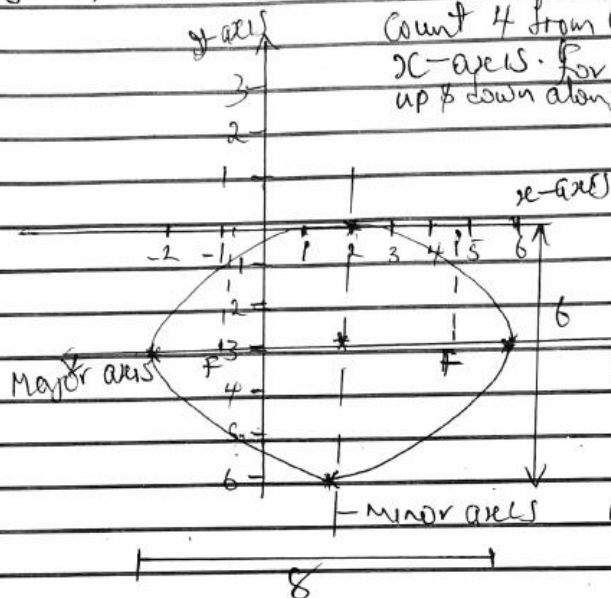
Soln

$$a^2 = 16 \Rightarrow a = 4$$

$$b^2 = 9 \Rightarrow b = 3$$

Centre $C_c(2, -3) \Rightarrow h = 2 \text{ \& } k = -3$

To draw ellipse. From centre $a = 4$
Count 4 from centre both right & left on
x-axis. For $b = 3$ count 3 from centre
up & down along y-axis



$$\text{Major axis} = 2a = 8$$

$$\text{Minor axis} = 2b = 6$$

$$\text{Vertex } V(h \pm a, k)$$

$$V(2 \pm 4, -3)$$

$$(6, -3) \text{ \& } (-2, -3)$$

on major axis

$$\text{Vertices on minor axis } V(h, k \pm b) \Rightarrow (2, -3 \pm 3)$$

$$(2, 0) \text{ \& } (2, -6)$$

$$\text{To find foci } c^2 = a^2 - b^2 \Rightarrow c^2 = 16 - 9 \Rightarrow c^2 = 7$$

$$F(h \pm c, k)$$

$$c = \sqrt{7}$$

$$\text{foci } (2 \pm c, k) \Rightarrow f(2 \pm \sqrt{7}, -3) = (2 \pm 2.65, -3)$$

$$= (4.65, -3) \text{ \& } (-0.65, -3)$$

Do not
write in
this margin

QUESTION NO.

Do not
write in
this margin

Example 3 $\frac{(x+1)^2}{9} + \frac{(y-2)^2}{25} = 1$

Soln

$$a > b \quad \therefore a^2 = 25 \Rightarrow a = 5$$

$$b^2 = 9 \Rightarrow b = 3$$

$$C(h, k) = C(-1, 2) \text{ — centre}$$

$$\text{Vertices, } V(h, k \pm a) = V(-1, 2 \pm 5)$$

$$V(-1, 7) \text{ \& } V(-1, -3)$$

$$\text{Focal, } F(h, k \pm c)$$

$$c^2 = a^2 - b^2$$

$$= 5^2 - 3^2$$

$$c^2 = 16$$

$$c = 4$$

$$F(-1, 2 \pm 4)$$

$$F(-1, 6) \text{ \& } F(-1, -2)$$

Vertices

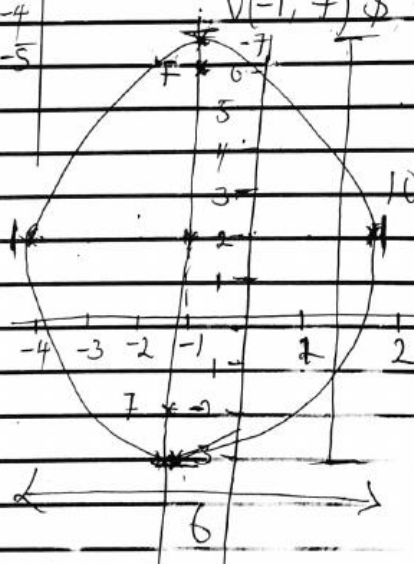
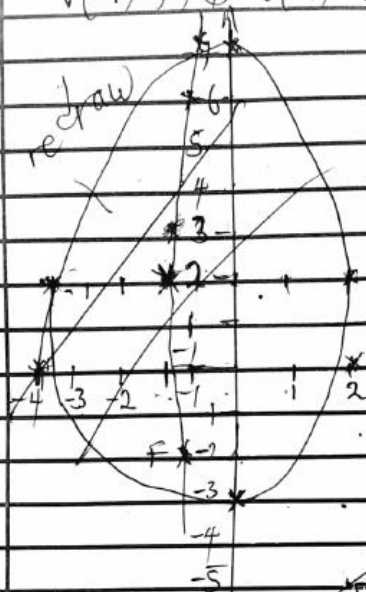
$$V(h, k \pm a)$$

$$V(-1, 2 \pm 5)$$

$$V(-1, 7) \text{ \& } V(-1, -3)$$

$$\text{Major axis} = 2a = 10$$

$$\text{minor axis} = 2b = 6$$



Hyperbola

Do not
write in
this marginDo not
write in
this margin

Hyperbola QUESTION NO.....

Standard Equation

(a) $\frac{(x-h)^2}{a^2} - \frac{(y-k)^2}{b^2} = 1$ where a & b are constant there/4 the major axis lies on x -axis
Centre, $C(h, k)$

(b) $\frac{(y-k)^2}{a^2} - \frac{(x-h)^2}{b^2} = 1$ where a & b are constant there/4 the major axis lies on y -axis

Vertices are $V(h, k \pm a)$ Focal, $F(h, k \pm c)$ where $c^2 = a^2 + b^2$ Major axis, $MA = 2a$ Minor axis $= 2b$

Ex: $\frac{x^2}{9} - \frac{y^2}{4} = 1$

$a^2 = 9, b^2 = 4$

Centre, $C(0, 0)$ i.e. $h=0, k=0$

$\therefore a = 3$ & $b = 2$

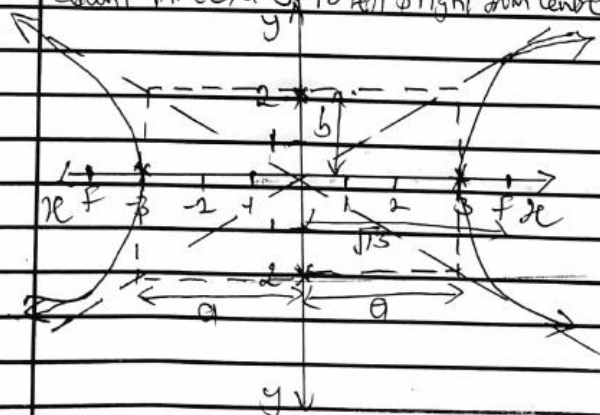
$c^2 = a^2 + b^2$

$c^2 = 9 + 4 = 13$

$c = \sqrt{13}$

Focal $F(h \pm c, k)$

$F(0 \pm \sqrt{13}, 0)$ or $(3, 0)$ & $(-3, 0)$

Count three, $a=3$, to left & right from centre

Asymptotes

$y - k = \pm \frac{b}{a}(x - h)$

Compare with

$y = mx + b$

$y = \pm \frac{2}{3}x$

Vertex of Major $V(h \pm a, k)$

$V(0 \pm 3, 0) = V(\pm 3, 0)$ i.e. $V(3, 0)$ & $V(-3, 0)$

Do not
write in
this margin

QUESTION NO.....

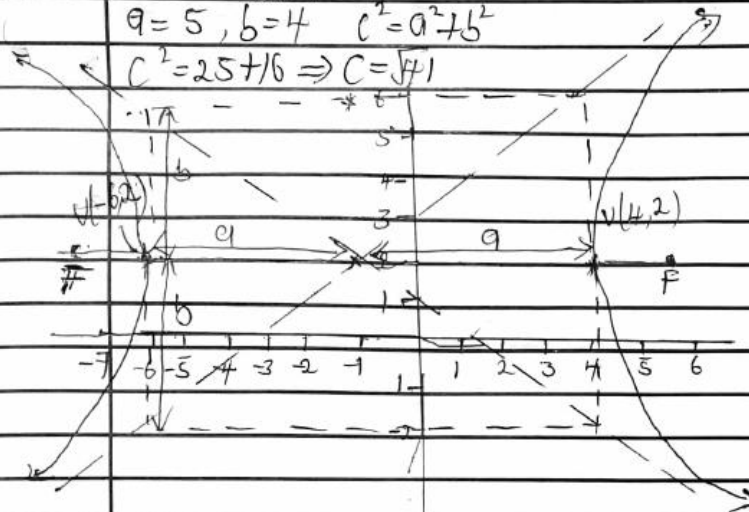
Do not
write in
this margin

Ex $\frac{(x+1)^2}{25} - \frac{(y-2)^2}{16} = 1$

Soln

$$a=5, b=4 \quad c^2 = a^2 + b^2$$

$$c^2 = 25 + 16 \Rightarrow c = \sqrt{41}$$



$$V(h \pm a, k) = V(-1 \pm 5, 2)$$

$$V(4, 2) \text{ \& } V(-6, 2)$$

$$\text{Foci, } F(h \pm c, k) = F(-1 \pm \sqrt{41}, 2)$$

$$F(-1 \pm 6.4, 2) = F(5.4, 2) \text{ \& } F(-7.4, 2)$$

Asymptotes

$$y - k = \pm \frac{b}{a}(x - h)$$

$$y - 2 = \pm \frac{4}{5}(x + 1)$$

Do not
write in
this margin

QUESTION NO.

Do not
write in
this margin

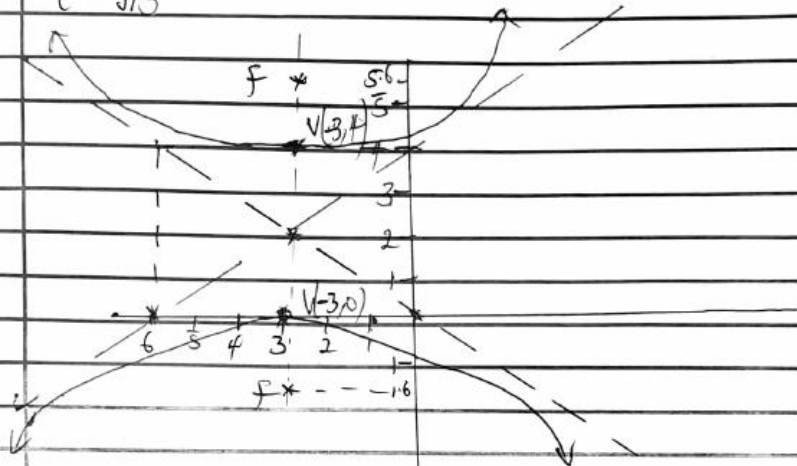
Example $\frac{(y-2)^2}{4} - \frac{(x+3)^2}{9} = 1$

Soln

$$a=2, b=3, \text{ Centre } (-3, 2)$$

$$c^2 = a^2 + b^2 = 4 + 9$$

$$c = \sqrt{13}$$



$$f(h, k \pm c)$$

$$f(-3, 2 \pm \sqrt{13})$$

$$f(-3, 2 \pm 3.6) = f(-3, 5.6) \text{ \& } f(-3, -1.6)$$

$$v(h, k \pm a) = v(-3, 2 \pm 2)$$

$$v(-3, 4) \text{ \& } v(-3, 0)$$

Asymptotes

$$y - k = \pm \frac{a}{b}(x - h)$$

$$y - 2 = \pm \frac{2}{3}(x + 3)$$

Paper 2 of 5 — Cyssa hub100LVL materials2nd SemesterMTH 104IMG-20250826-WA0000.jpg

Lecture/Notes · Google Drive harvest

Paper 3 of 5 — Cyssa hub100LVL materials2nd SemesterMTH 104IMG-20250826-WA0001.jpg

Lecture/Notes · Google Drive harvest

Paper 4 of 5 — Cyssa hub100LVL materials2nd SemesterMTH 104IMG-20250826-WA0002.jpg

Lecture/Notes · Google Drive harvest

Paper 5 of 5 — Cyssa hub100LVL materials2nd SemesterMTH 104IMG-20250826-WA0003.jpg

Lecture/Notes · Google Drive harvest
